ASEAN GEOAI FUSION 2026



Dr. Fathoni Usman, M.Eng., Eng.Tech., IPM, ASEAN Eng.
UNITEN

Architecting GeoAI Systems on the Cloud

Platform Selection, Telecom Applications, and Cost Optimization Frameworks

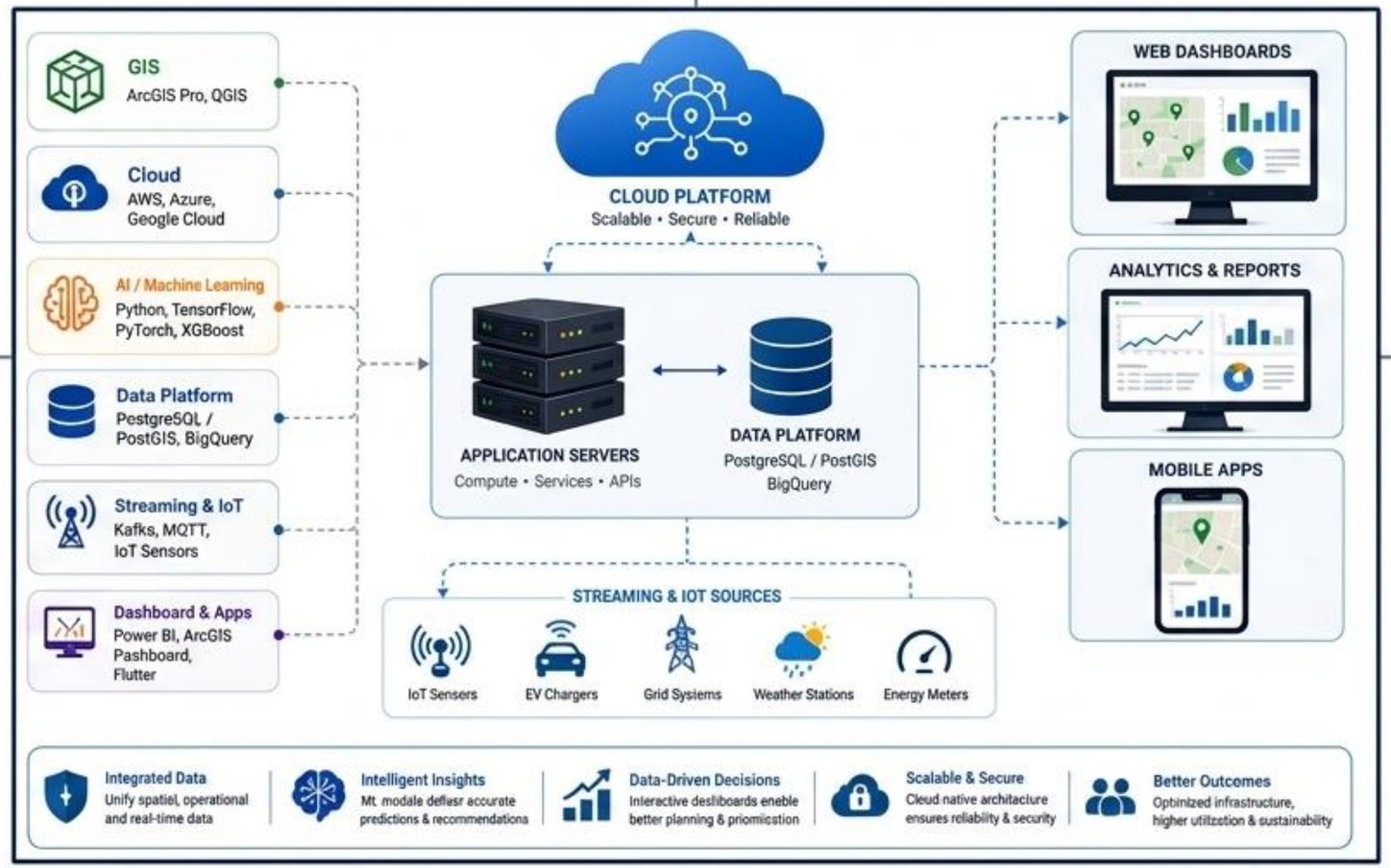
Integrating Geospatial Intelligence with Scalable Cloud Infrastructure

ASEAN
GEOAI FUSION 2026

The Central Nervous System

Scalable, secure cloud infrastructure is the non-negotiable core of modern GeoAI.

Managed compute clusters process heavy raster/vector models.

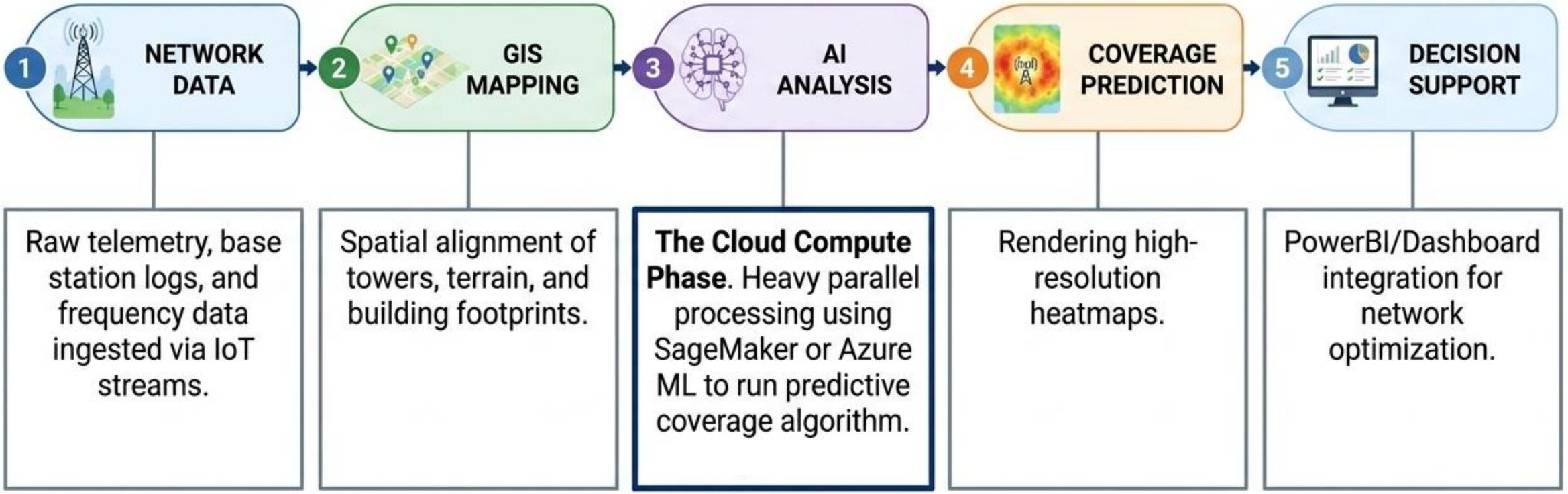


Spatial databases rely on cloud-backed durability and high-throughput IOPS.

The GeoAI stack requires a robust, scalable cloud backend to translate raw spatial and IoT data into actionable intelligence.

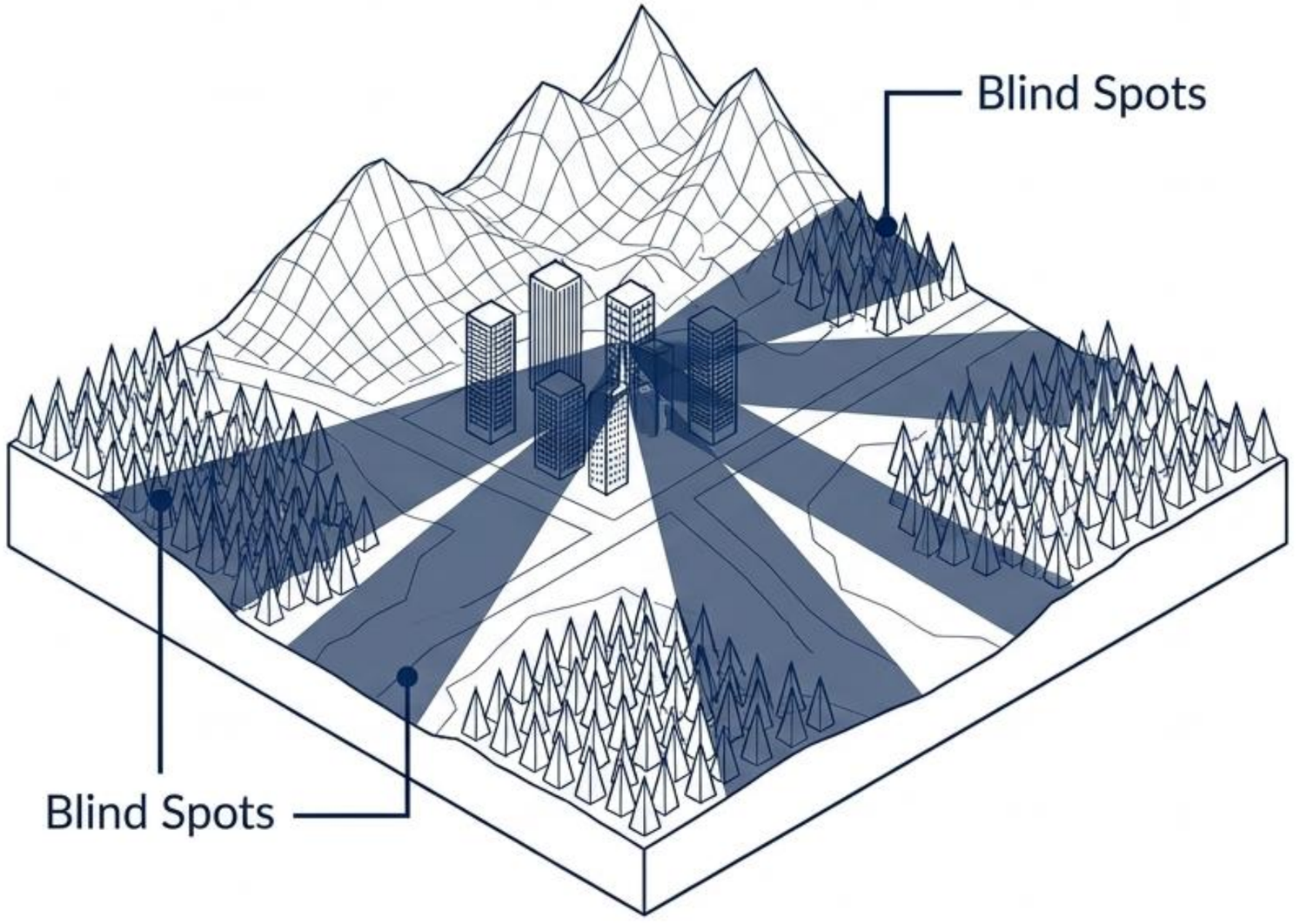
Defining the Cloud Workload: Telecom Network Coverage

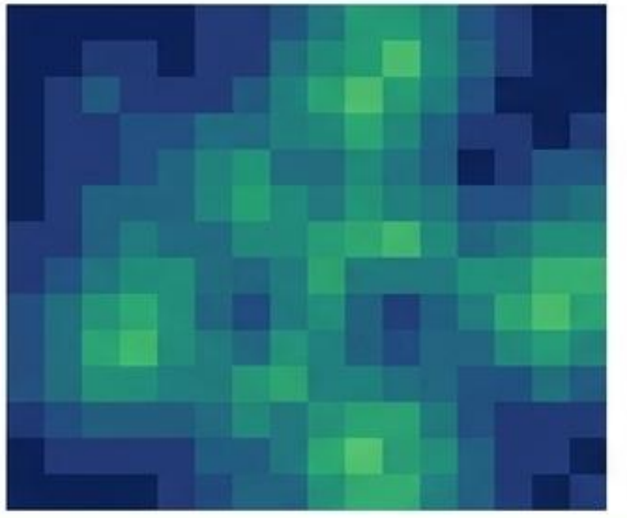
ASEAN GEOAI FUSION 2026



Cloud scalability is non-negotiable at Phase 3—where massive, multi-variable spatial datasets meet deep learning algorithms.

Use Case I: Telecom Hazard Response & Network Resilience





The Problem

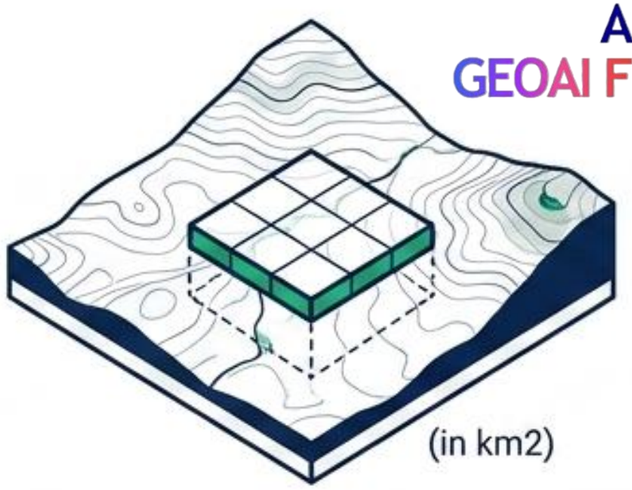
Natural hazards disrupt telecom networks, delaying rescues.

The GeoAI Solution

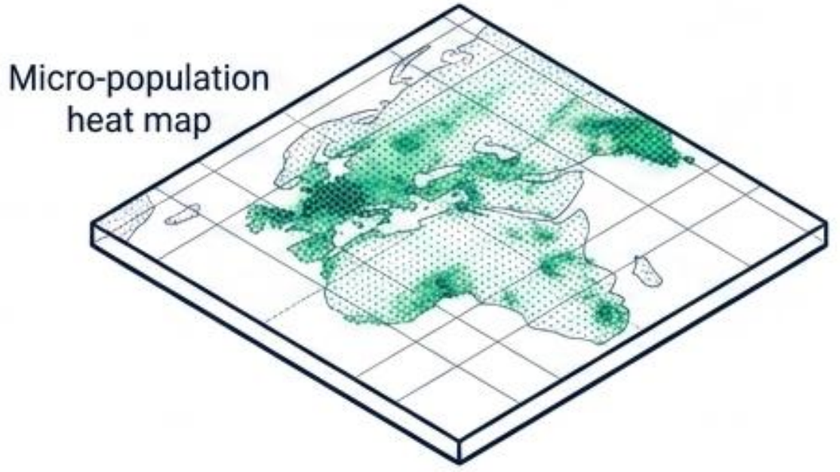
Predict impact, prioritize actions, and optimize recovery.

The Analytical Engine: Population Coverage Estimation

ASEAN GEOAI FUSION 2026



$$P_{covered} = \sum_{i=1}^{n} (Area_i \times Density_i \times C_i)$$





Binary Coverage Factor
(1 = Covered; 0 = Dead zone)

GeoAI prioritizes bandwidth allocation where people actually congregate, closing the digital divide.

Use Case II: EV Charging Infrastructure Planning

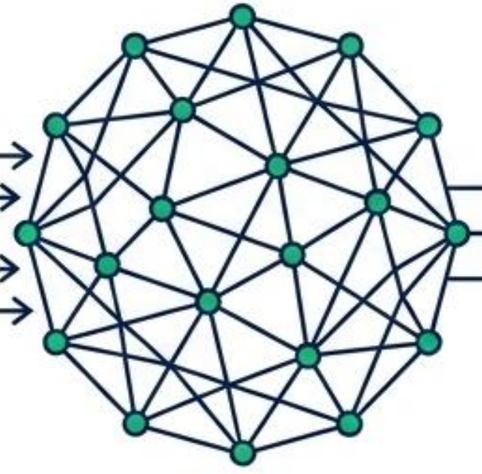
Simulate & Decide:
What happens if we build 50 new chargers?

Road Network

Substations

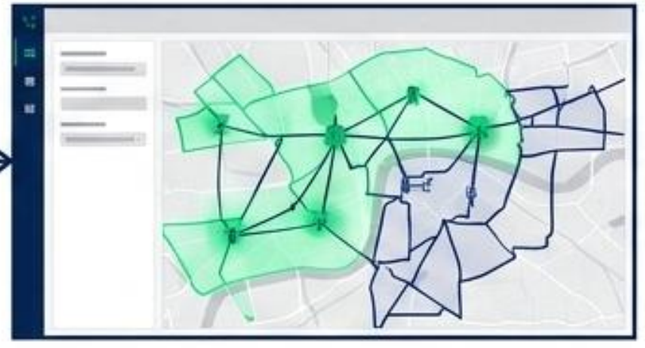
EV Registrations

Income Levels

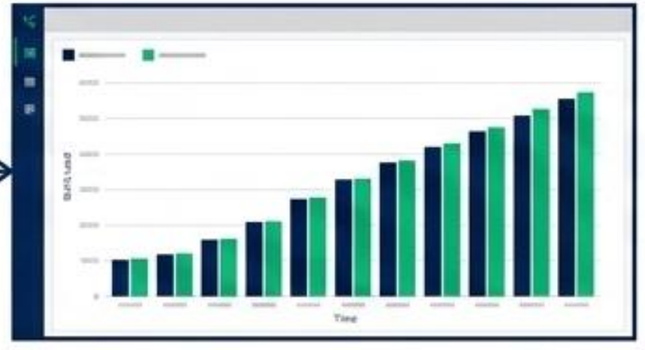




Accessibility modelling



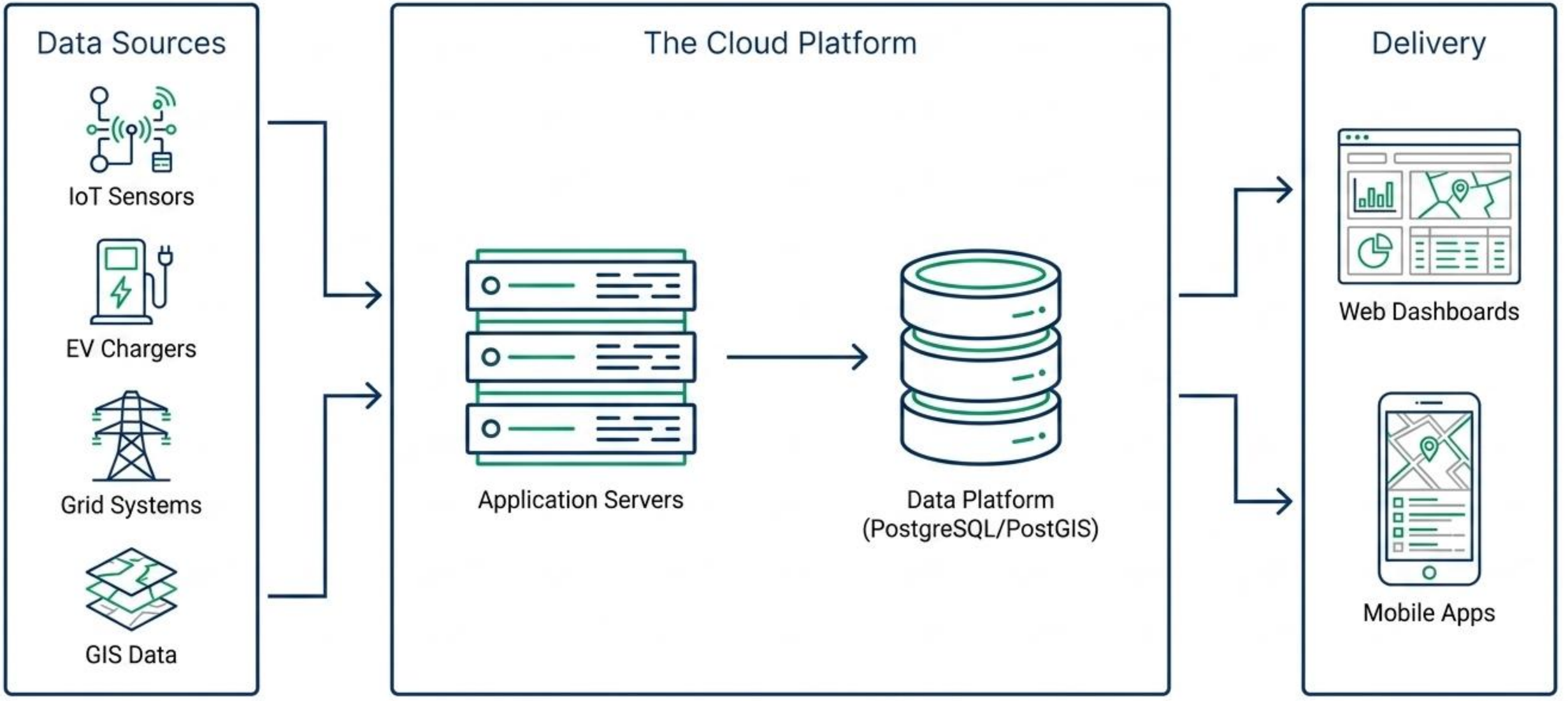
Grid capacity prediction



Site recommendation engine

The Infrastructure Blueprint: From Sensor to Dashboard

Translating spatial intelligence into scalable, secure, cloud-native architecture.



Strategic Platform Matrix: AWS vs. Azure vs. Google

ASEAN GEOAI FUSION 2026

	AWS ECOSYSTEM	AZURE ECOSYSTEM	GOOGLE COLAB & GEE
Core Strength	Massive scale & Open Data	Enterprise BI integration	Rapid Prototyping & Research
Key Spatial Tools	Amazon Location Service, SageMaker	Synapse Spatial Analytics, Azure Maps	Earth Engine (Petabyte satellite archive)
Compute Paradigm	Heavy GPU clusters, S3 Data Lakes	Unified data warehouse queries	Interactive Python/Jupyter environments
Primary User Base	Production engineers handling petabytes	Orgs locked into Microsoft tools	Academic professionals, data scientists

The Prototyping Engine: Google Earth Engine & Colab



Google Earth Engine (GEE)

Instant access to petabytes of historical satellite imagery and geospatial datasets without managing underlying storage.

API Node Integration

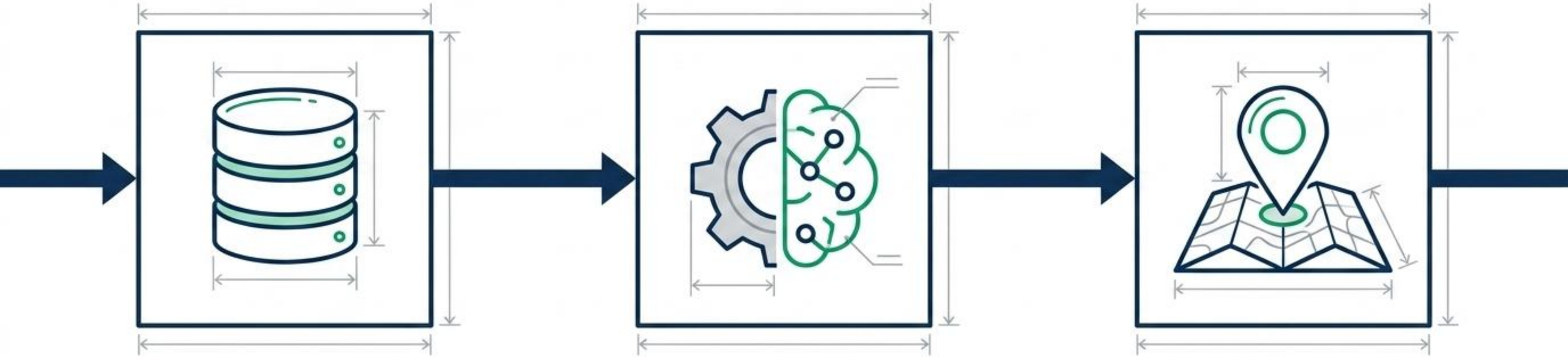


Google Colab

The zero-friction Python execution environment. Enables academic professionals and innovators to build ML models directly in the browser.

The ideal hub for bootcamps, hackathons, and certified assignments (e.g., uploading to the Colab Hub).

Production Scale: The AWS Geospatial Ecosystem



S3 + Open Data

Hosts Landsat and Sentinel archives at no storage cost, lowering the barrier for massive imagery processing.

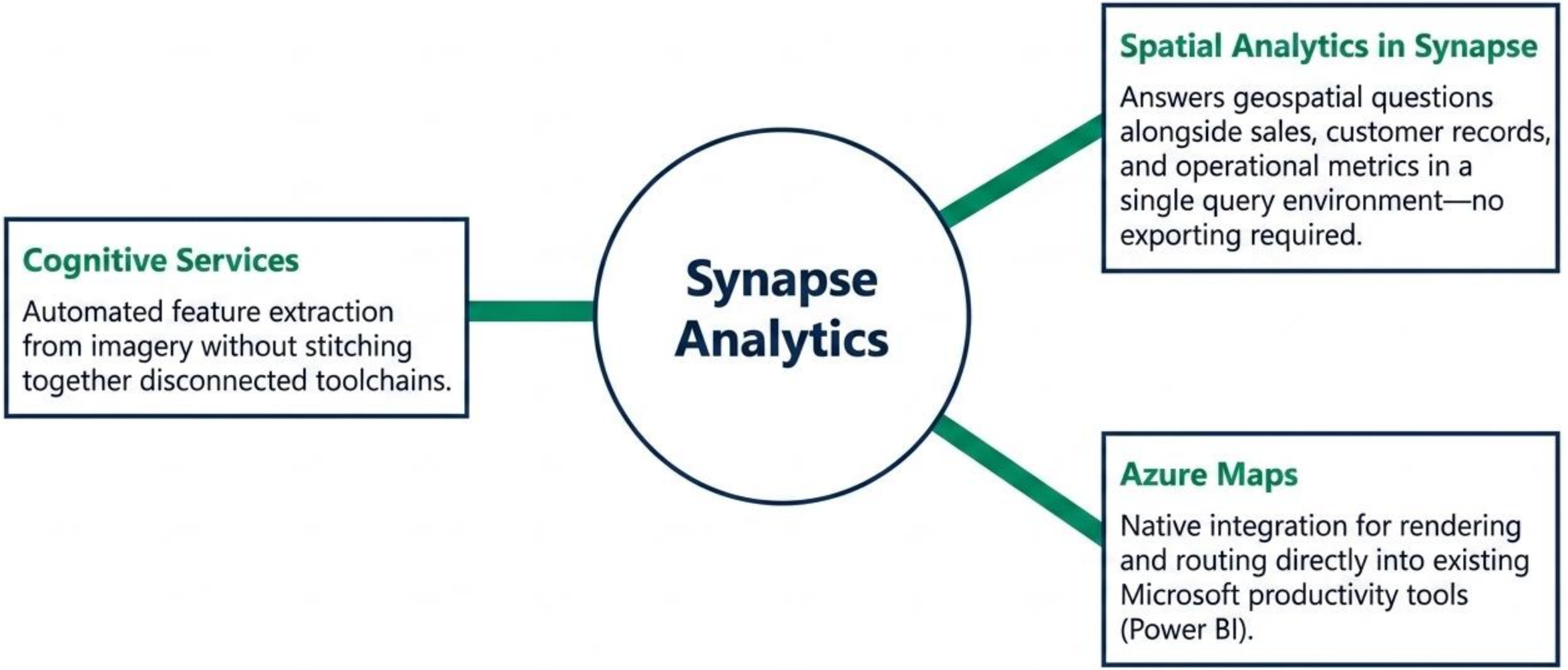
SageMaker

Moves raw satellite data to deployed prediction models without the user managing custom GPU clusters.

Location Service

Managed maps, routing, and geocoding APIs without negotiating with external map providers.

Enterprise Integration: The Azure Spatial Ecosystem



Strategic Platform Selection Criteria

Scale of Data

Is this a pilot project (Colab/GEE) or a petabyte daily pipeline (AWS)?
Match infrastructure to actual scale to avoid over-provisioning.

Existing Ecosystem

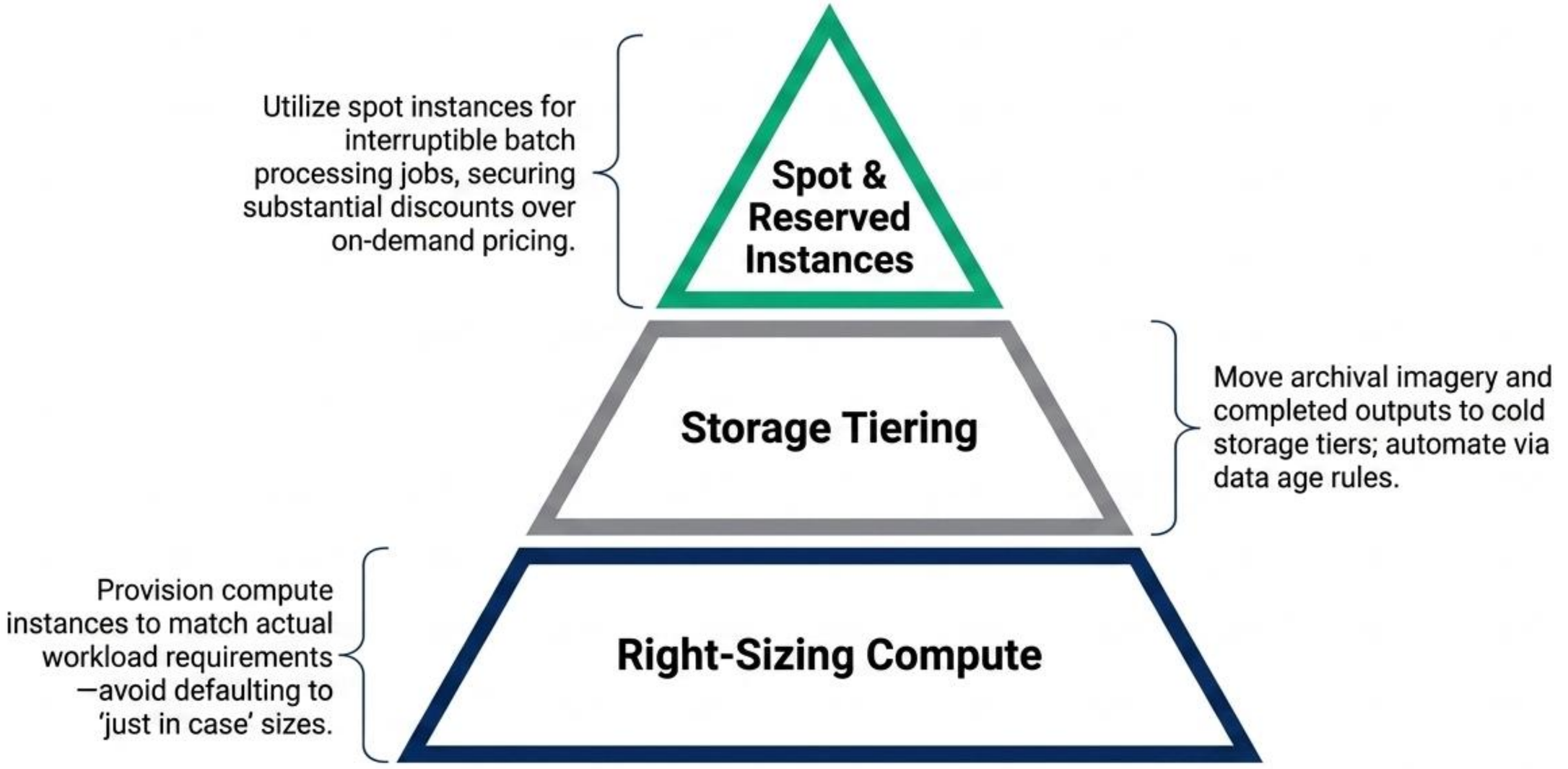
Are you already utilizing AWS data warehouses or Azure identity management?
Stay within your ecosystem to limit compound integration friction.

Skill Availability

The most technically capable platform is not necessarily the right choice if the team lacks the expertise to operate it confidently.

GeoAI FinOps: Cloud Cost Optimization

ASEAN GEOAI FUSION 2026



Delivering Measurable Impact: The Ultimate GeoAI Goal



Economic

Lower downtime, optimized operations, increased ROI for telecom and infrastructure investments.



Social

Improved public safety, faster disaster communication restoration, and bridging the digital divide.



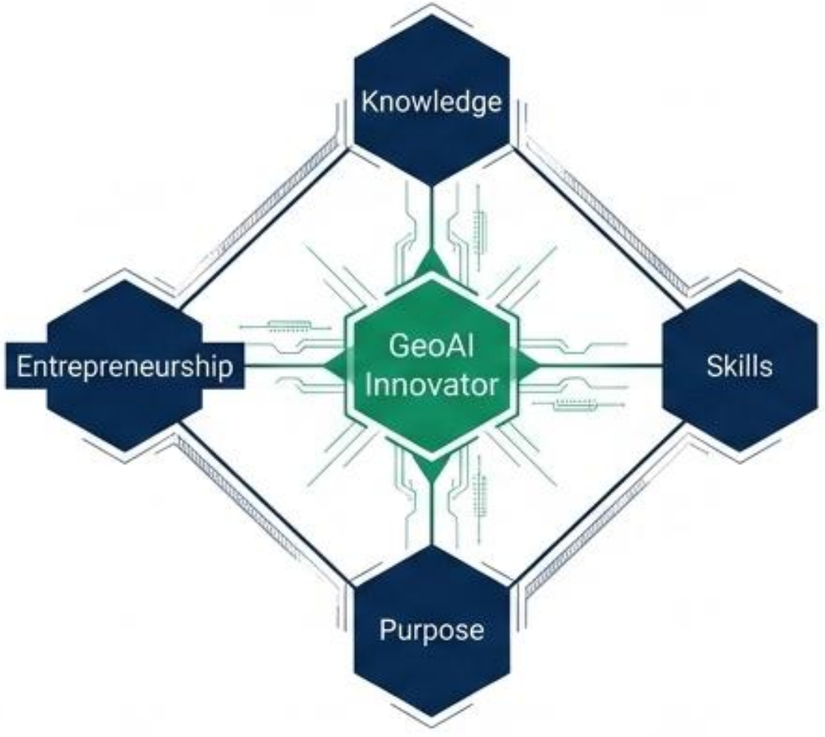
Environmental

Smarter resource use, lower carbon footprints, and climate-resilient digital infrastructure.

Stronger Networks. Faster Response. Safer Communities.
GeoAI for a Connected & Resilient ASEAN.

Dual Frameworks: Developing the Innovator and the Innovation

The Human: GeoAI Innovator Canvas



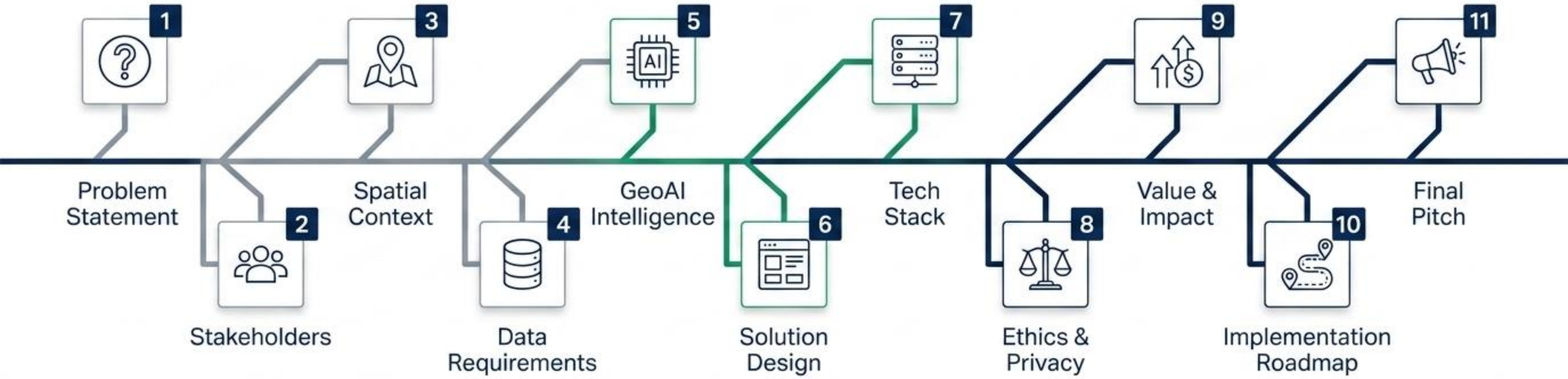
The Product: PDGS Canvas



The GIC Canvas is about building the GeoAI Innovator;
The PDGS Canvas is about building the GeoAI Innovation.

The 11-Step PDGS Transformation Pipeline

Systematically transforming a real-world problem into a responsible, implementable, and impactful GeoAI Solution.



The GeoAI Innovation Canvas: Sustainable Commitment

DIMENSION 2 – SUSTAINABLE COMMITMENT

What sacrifices are you making to sustain and complete the innovation until final Hackathon?

SUSTAINABLE COMMITMENT

What sacrifices are you making to sustain and complete the innovation until final Hackathon?



> Innovation requires sacrifice today for a greater impact tomorrow.

SACRIFICE AREA	ONE LINE EXAMPLES OF SACRIFICES
Time Commitment	I sacrifice my free time, weekends, and holidays to work on this GeoAI solution.
Personal & Family Time	I sacrifice time with my family and friends to focus on building and improving the innovation.
Financial Resources	I invest my own money for data access, tools, cloud credits, travel, and other project needs.
Academic / Work Priorities	I postpone or reduce other assignments, research, or work tasks to prioritise this innovation.
Energy & Well-being	I sacrifice sleep, rest, and personal comfort to meet milestones and improve our solution.
Other Opportunities	I turn down other opportunities and activities to stay focused on completing the innovation.
Consistent Effort	I commit to consistent practice, learning, and team collaboration until the final Hackathon.
Long-term Commitment	I stay committed to seeing this innovation through, from ideation to final Hackathon.

Technology is only half the architecture; the rest is human commitment.

- **Time Commitment:** Sacrificing weekends and free time to build GeoAI solutions.
- **Financial Resources:** Investing personal or institutional capital for cloud credits, API access, and tools.
- **Academic/Work Priorities:** Postponing alternative research to prioritize geospatial innovation.

Small sacrifices today build resilient digital infrastructure for tomorrow.